

Lankireddy Vinith Reddy

 Github |  LinkedIn |  vinithlankireddy@gmail.com |  +91 7995660947
Vallabhapuram, Guntur, Andhra Pradesh

EDUCATION

2023 - 2027 B.Tech CSE (AI Specialization) at **Bennett University**, Uttar Pradesh (CGPA: 9.25/10)
2021 - 2023 Intermediate (MPC) at **FIITJEE Junior College**, Andhra Pradesh (91.9%)

TECHNICAL SKILLS

Languages	Python, Java, C++
ML/DL	PyTorch, TensorFlow, Scikit-learn, Keras, NLP (NLTK, Hugging Face)
Agentic AI	LangChain, LangGraph, RAG Pipelines, Vector DBs (Qdrant)
Data Handling	Pandas, NumPy, SQL
Backend	FastAPI
Tools	Git, GitHub, Docker

PROJECTS

Float Chat – Argo Float AI Query System | *LangGraph, LangChain, PostgreSQL, Flask* [GitHub Link](#)

- Built a multi-agent AI system to query and analyze Argo oceanographic float data using a Main Agent, Filter Agent, and SQL Agent orchestrated via LangGraph.
- Engineered a Filter Agent supporting geographic, temporal, parameter-based, and negation queries over float metadata with composite filter chaining.
- Developed an SQL Agent that translates natural language into PostgreSQL queries, retrieves depth-profile measurements, and returns structured JSON grouped by region.
- Designed the Main Agent as an orchestrator performing relevance classification, query decomposition, and end-to-end workflow management using LangGraph state machines.

Realtime AI Chat Backend | *FastAPI, LangChain, LangGraph* [GitHub Link](#)

- Developed a real-time conversational AI backend using FastAPI and WebSockets with token-level streaming.
- Built a LangGraph-based agent with tool calling and multi-turn reasoning via Google Gemini.
- Implemented persistent chat sessions, message storage, and post-session summaries using Supabase (PostgreSQL).
- Designed a responsive chat UI with live AI streaming and background task processing.

Real-Time Video Segmentation | *PyTorch, OpenCV, SAM 2, U-Net* [GitHub Link](#)

- Built a real-time video segmentation pipeline using SAM 2 for automatic mask generation and tracking.
- Developed a lightweight U-Net achieving high FPS for real-time inference.
- Achieved ~ 0.69 Validation Dice and ~ 0.97 Training Dice; tracked experiments with Weights & Biases.
- Conducted real-time inference and video blending with OpenCV, achieving continuous, low-latency tracking performance suitable for dynamic manipulation hardware.

ACHIEVEMENTS

- **Dean's Certificate for Academic Excellence (2024)** — Awarded for outstanding academic performance.

CERTIFICATIONS

- **Build Basic Generative Adversarial Networks (GANs) (2026)** [Verification Link](#)